

COMPUTER EXERCISE 1

VALENCIA 2019

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1. EXERCISE 1

In this exercise we will analyse the *savings* dataset from the *faraway* package:

```
> library(faraway)
> data(savings)
```

The following commands give us a glimpse at the data i.e. attributes of the data and their values, as well as univariate summary statistics of each attribute, for example, max, min, mean etc:

```
> head(savings)
```

	sr	pop15	pop75	dpi	ddpi
Australia	11.43	29.35	2.87	2329.68	2.87
Austria	12.07	23.32	4.41	1507.99	3.93
Belgium	13.17	23.80	4.43	2108.47	3.82
Bolivia	5.75	41.89	1.67	189.13	0.22
Brazil	12.88	42.19	0.83	728.47	4.56
Canada	8.79	31.72	2.85	2982.88	2.43

```
> summary(savings)
```

	sr	pop15	pop75	dpi
Min.	: 0.600	Min. :21.44	Min. :0.560	Min. : 88.94
1st Qu.:	6.970	1st Qu.:26.21	1st Qu.:1.125	1st Qu.: 288.21
Median :	10.510	Median :32.58	Median :2.175	Median : 695.66
Mean :	9.671	Mean :35.09	Mean :2.293	Mean :1106.76
3rd Qu.:	12.617	3rd Qu.:44.06	3rd Qu.:3.325	3rd Qu.:1795.62
Max. :	21.100	Max. :47.64	Max. :4.700	Max. :4001.89

	ddpi
Min.	: 0.220
1st Qu.:	2.002
Median :	3.000
Mean :	3.758

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```
3rd Qu.: 4.478
Max.    :16.710
```

A couple of things stand out. We can see a max reading under the attribute, `pop15`, of 47.64. Almost half of the total population is under the age of 15. This suggests (at least to me) that the country is probably a poor country. Similarly under the attribute, `pop75`, there is a min reading of 0.56 i.e. about half a percent of the total population are over 75 years of age. This is quite extreme and would suggest again that the country is probably poor.

A histogram and density plot of the savings rate shows a skewed left distribution. We should expect to see this given that the mean is less than the median under the summary stats for that variable.

```
> par(mfrow = c(1,2))
> hist(savings$sr, xlab = "Savings Rate", main = "")
> plot(density(savings$sr, na.rm = TRUE), main = "")
```

Below are plots of sorted data against its index for both the variable, `sr` and `pop15`. These reveal, respectively, that there may be a couple of outliers (at the top end of the indices) in the variable, `sr`, and for the variable, `pop15`, two groups in the distribution seem to show up. Investigating this a little bit further, we see that there are two distinct groups in the scatter plot of `sr` vs `pop15`. This would suggest some kind of dependence between age and amount of savings - which makes sense. The plot seems to show that those countries with a smaller proportion of under 15's have a higher `sr`. Again this makes sense.

```
> par(mfrow = c(1,3))
> plot(sort(savings$sr), ylab = "Sorted Savings Rate")
> plot(sort(savings$pop15), ylab = "Sorted Perc. of Pop < 15")
> plot(sr ~ pop15, savings)
>
```

The following code fits a linear model to the data and plots the variable, `sr`:

```
> lmod1 <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data = savings)
> plot(savings$sr, type = 'n')
> for (i in 1:length(savings$sr)) {
+   text(x = i, y = savings$sr[i],
+        labels = (rownames(savings)[i]), cex = 0.5)
+ }
```

A summary of the data follows:

```
> summary(lmod1)
```

Call:

```
lm(formula = sr ~ pop15 + pop75 + dpi + ddpi, data = savings)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-8.2422	-2.6857	-0.2488	2.4280	9.7509

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28.5660865	7.3545161	3.884	0.000334 ***
pop15	-0.4611931	0.1446422	-3.189	0.002603 **
pop75	-1.6914977	1.0835989	-1.561	0.125530
dpi	-0.0003369	0.0009311	-0.362	0.719173
ddpi	0.4096949	0.1961971	2.088	0.042471 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.803 on 45 degrees of freedom

Multiple R-squared: 0.3385, Adjusted R-squared: 0.2797

F-statistic: 5.756 on 4 and 45 DF, p-value: 0.0007904

We get a significant readings for both `pop15` and `ddpi`. The former means that, all else being the same, a unit increase in the proportion of under 15's results in almost half a unit decrease in savings. This agrees with what we saw pictorially. The latter significant reading says that a unit increase in `ddpi` results in just under half a unit increase in savings. Again this makes sense since the faster your disposable income increases the more you will have to save.

The constant variance and normality assumption of the residuals seem to hold up looking at top two plots below. However, the third plot suggests there is dependence of the residuals between countries with high or low proportion of under 15's. Furthermore, within each group we can see clusters of observations above and below the line $\hat{\epsilon} = 0$. Thus the independence assumption is violated.

```
> par(mfrow=c(2,2))
> plot(fitted(lmod1), residuals(lmod1), xlab = "Fitted",
+      ylab = "Residuals") # const variance
> abline(h=0)
> # normality
> qqnorm(residuals(lmod1), ylab = "Residuals", main="")
> qqline(residuals(lmod1))
> # correlation
```

```
> plot(residuals(lmod1) ~ pop15, ylab = "Residuals", savings)
> abline(h=0)
```

2. EXERCISE 2

In this exercise we consider the effect of removing Sweden and Libya, one at a time, on the fit of the model.

```
> lmod1_adj1 <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data = savings,
+                 subset = (rownames(savings) != "Sweden"))
> lmod1_adj2 <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data = savings,
+                 subset = (rownames(savings) != "Libya"))
```

The summaries of the above fits with the original are given below:

```
> sumary(lmod1)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28.56608654	7.35451611	3.8842	0.0003338
pop15	-0.46119315	0.14464222	-3.1885	0.0026030
pop75	-1.69149768	1.08359893	-1.5610	0.1255298
dpi	-0.00033690	0.00093111	-0.3618	0.7191732
ddpi	0.40969493	0.19619713	2.0882	0.0424711

n = 50, p = 5, Residual SE = 3.80267, R-Squared = 0.34

```
> sumary(lmod1_adj1)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.82712563	7.34399157	3.7891	0.0004558
pop15	-0.44944520	0.14426001	-3.1155	0.0032284
pop75	-1.62501060	1.07967187	-1.5051	0.1394446
dpi	-0.00010038	0.00094715	-0.1060	0.9160790
ddpi	0.41229457	0.19524173	2.1117	0.0404251

n = 49, p = 5, Residual SE = 3.78392, R-Squared = 0.35

```
> sumary(lmod1_adj2)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	24.52404598	8.22402631	2.9820	0.004655
pop15	-0.39144013	0.15790949	-2.4789	0.017084
pop75	-1.28086692	1.14518206	-1.1185	0.269430
dpi	-0.00031890	0.00092933	-0.3432	0.733119
ddpi	0.61027903	0.26877842	2.2706	0.028122

n = 49, p = 5, Residual SE = 3.79481, R-Squared = 0.36

Looking at the above summaries we can see that by removing Sweden from the dataset nothing really changed. However, removing Libya

from the dataset effect the fit of the model. Indeed, for the variable, `sr`, there is an increase of 16.362% in its estimated value and for the variable, `ddpi`, the estimate also increases but by 39.331%. We would consider this an influential observation.

3. EXERCISE 3

This exercise just involves generating the following plot:

```
> plot(savings$sr ~ savings$pop15, type='n')
> for (i in 1:length(savings$sr)) {
+   text(x = savings$pop15[i], y = savings$sr[i],
+        labels = (rownames(savings)[i]), cex = 0.5)
+ }
```

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